

Hackathon – Family Asset Tracker

Track personal and family assets in one place with an offline-first, privacy-focused experience.

Product goal

Build a simple family asset tracker where members can record what they own, track Buy and Sell activity, understand personal and family-level asset allocation, and control who can see their profile.

Core principle: all members have the same access. There is no admin role.

1. Product Overview

Let's build a family asset tracker that brings personal assets and family assets into one place, while giving each person control over profile and holding visibility.

The app is offline-first and privacy-first. Data should come from a local database, and no API integration is required for the hackathon.

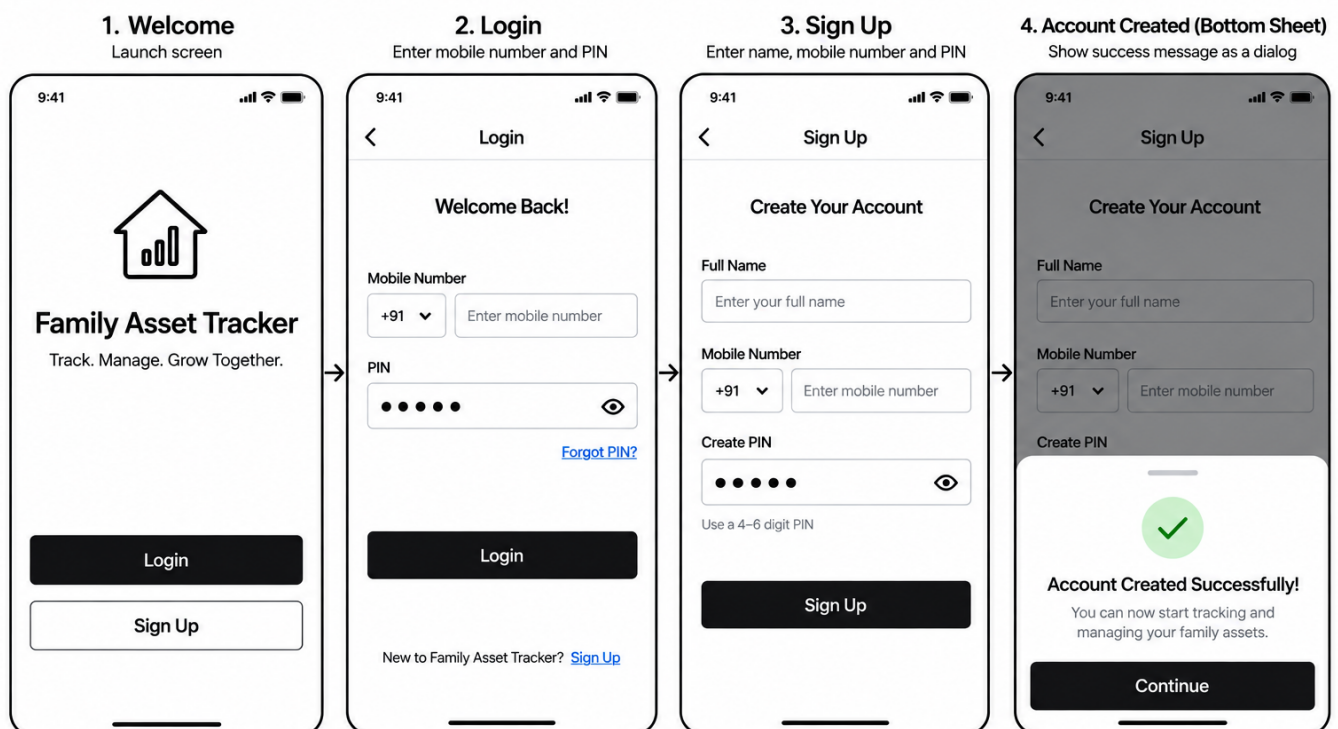
A holding represents an asset owned by a person. Assets can belong to categories such as Equity, Mutual Fund, Real Estate, Gold, Crypto and Other. The tracker focuses on two transaction actions: Buy and Sell.

2. Login & Sign Up

Users sign up with their name, mobile number and a 4–6 digit PIN. From the next visit, they can log in using the registered mobile number and PIN. Forgot PIN is outside the hackathon scope.

Login & Sign Up Flow

Family Asset Tracker – Login & Sign Up Flow



3. Key Use Cases

Family financial visibility: View total family assets and category allocation to understand concentration and make better decisions.

Guidance for children: Parents can use shared visibility of purchases and sales to understand how children are building their assets and provide guidance over time.

Filtering: A member can view another member's Equity purchases from the last year or Real Estate purchases across all time.

4. Core Data & Relationships

The app should maintain information about family members, their holdings, the different asset types and the assets or items being tracked. Each holding should support the transaction activity needed to represent what was bought or sold.

High-level relationship: Person → Holding → Asset Type → Asset → Buy / Sell

This relationship should support personal and family-level views, current asset values, historical changes, allocation charts and filtering. The candidate is expected to derive the appropriate data model, entities, attributes and relationships from the product requirements and user flows.

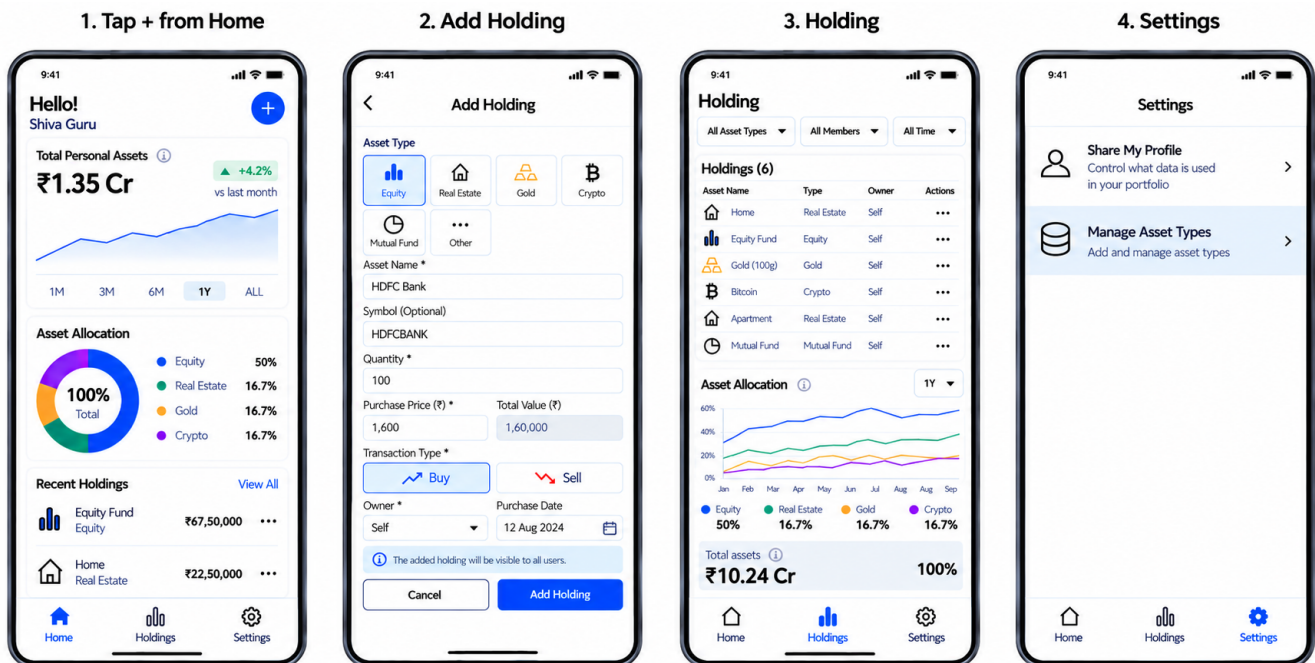
5. Home Screen

After login, the user lands on Home. Home represents the logged-in user's personal assets, not the family total.

Home has three sections: personal net worth with a mandatory line chart; an asset allocation visual showing percentage to total; and a list of recent holdings.

Reference data: personal total assets are ₹1.35 Cr. Equity represents 50%; the other displayed categories are equally split.

Home / Add Holding / Holding / Settings reference



6. Holding Screen

The Holding screen presents assets at the family level. The holding list can be filtered by Asset Type, Member and Time.

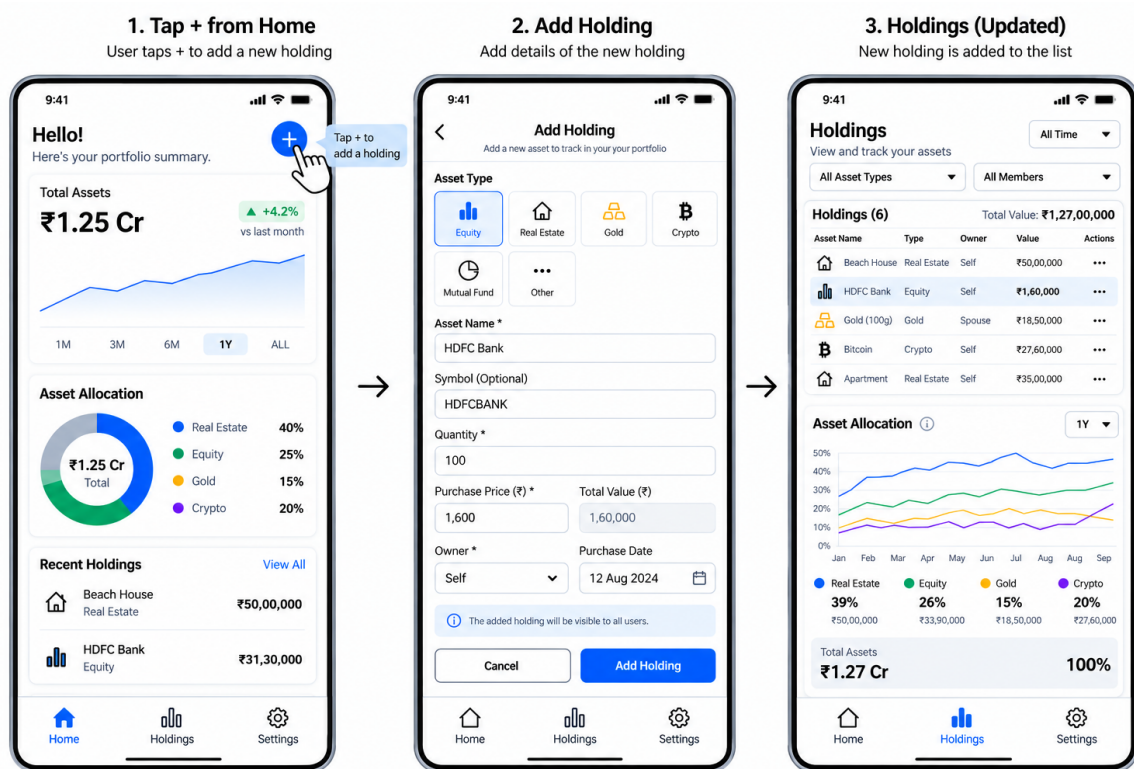
Keep the three filters grouped together in a compact flex-wrap layout: All Asset Types · All Members · All Time. The default is all asset types, all members and all time.

Below the holding list, a mandatory line chart shows family asset allocation over time. Under the chart, show only each category's percentage of total assets; actual category values are not

required.

The bottom family summary is fixed and shows Total assets: ₹10.24 Cr.

Holding screen reference



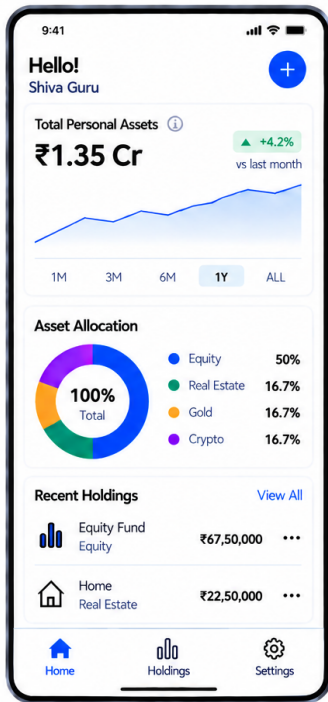
7. Add Holding

Users can record holdings for themselves or family members. The form must support both Buy and Sell transactions.

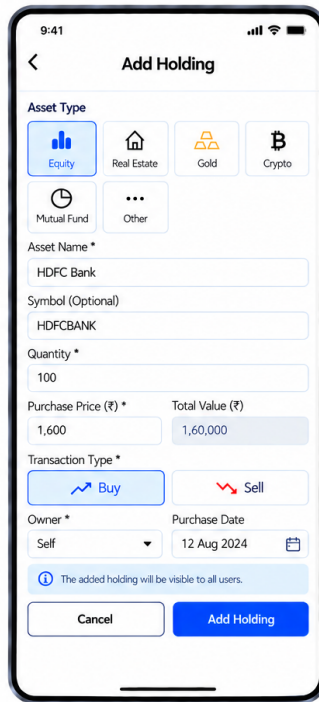
Capture the asset information, quantity/value, transaction type, owner and relevant date. The new holding should flow into the appropriate personal/family views and respect visibility settings.

Add Holding reference

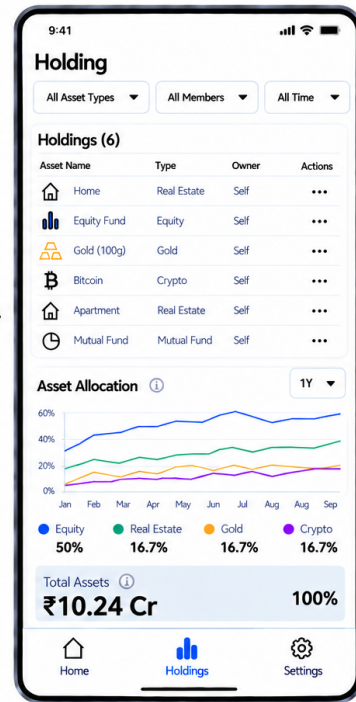
1. Tap + from Home



2. Add Holding



3. Holding



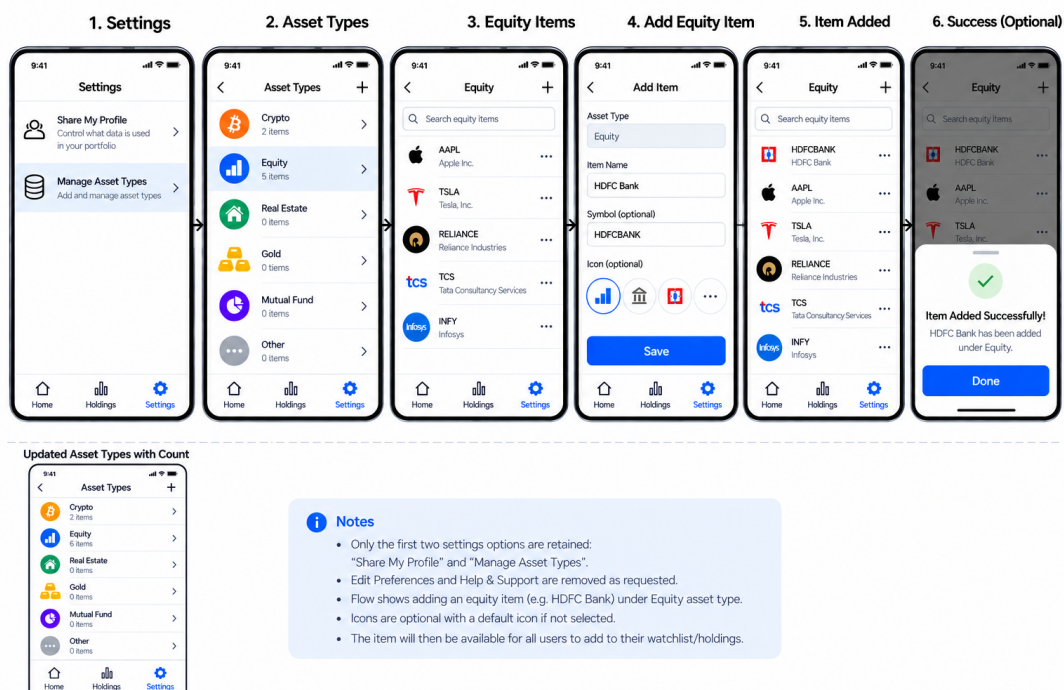
8. Settings

The mandatory Settings options are Share My Profile and Manage Asset Types.

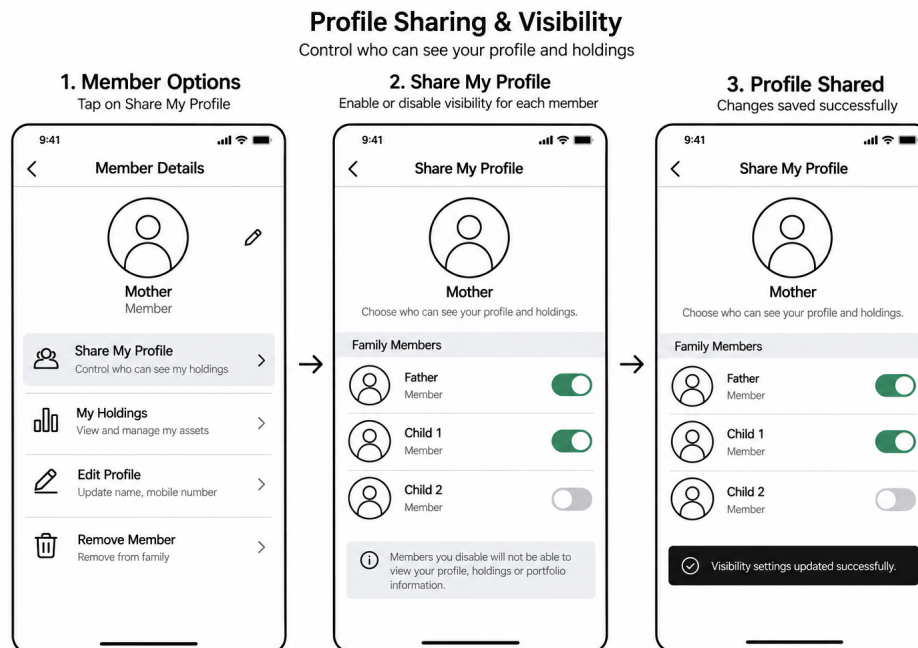
Share My Profile: by default, a person's assets are visible to all family members. The user can enable or disable visibility for individual members.

Manage Asset Types: optional for the hackathon. The preferred approach is to seed required asset types/items directly in the local database and focus on the functional core flow.

Settings / Manage Asset Types reference



Share My Profile / Visibility reference



9. Permission & Privacy Model

All registered family members have the same permissions and access. There is no admin concept. A person's assets are visible to all family members by default, while Share My Profile provides member-level visibility control.

10. Offline & Implementation Scope

- Use a local database as the source of truth.
- It is acceptable to seed the required data directly in the database.
- Screens should be functional; API integration is not required.
- Asset type/item management is optional and can be implemented after the core flows.

Charts: a doughnut chart is not mandatory as a specific chart type; any clear visual can communicate personal percentage-to-total. The line chart is mandatory on both Home and Holding.

11. Hackathon Candidate Focus

Focus on deriving the local data model and building the core experience end-to-end from the requirements and reference flows.

A successful implementation should demonstrate local login/sign-up, personal Home, family Holding with filters, Add Holding with Buy/Sell, visibility controls, calculations and mandatory charts.

Optional: asset type/item management and watchlist functionality.